



Anonymous function: Point Mass Kinematics

When a stone is thrown into the air with an angle θ_0 and a velocity v_0 relative to the horizon, it will have a horizontal component of velocity, v_x , and a vertical component v_y . If air resistance is ignored, the horizontal component remains constant, while the vertical component decreases due to gravity. An object falling under the influence of gravity on the Earth's surface accelerates at approximately $g = 9.81 \text{ m/s}^2$.

$$v_x = v_0 \cos(\theta_0)$$
$$v_y(t) = v_0 \sin(\theta_0) - gt$$

Write an anonymous function that takes the current time, t , as input and outputs the current vertical velocity, $v_y(t)$.

The anonymous function must be assigned to a variable called "func".

Script

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

Open Item in MATLAB Online



```
12 %%%
13
14
15 dt = 0.01;
16 t = 0:dt:8; % time vector
17
18 for i = 1:length(t)
19     vy(i) = func(t(i));
20 end
21
22 yline(vx, "--r")
23 hold on
24 plot(t, vy)
25 legend(["v_x", "v_y"])
26 ylabel("Velocity [m/s]")
27 xlabel("Time [s]")
28
```

Run Script

Tests: All Tests Passed

Submit

Evaluate anonymous function

Output

